MIL-P-18515B
11 May 1979
SUPERSEDING
MIL-P-18515A(OS)
16 March 1976 and
AMENDMENT 1
17 March 1978

MILITARY SPECIFICATION

PRIMER, PERCUSSION, MK 41 MOD 0

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the fabrication, preparation for delivery, and acceptance of the Percussion Primer, Mk 41 Mod 0, hereinafter referred to as the primer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501) Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-453	Inspection, Radiographic
MIL-STD-1167	Ammunition Data Cards

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

DL 563478	Primer, Percussion, Mk 41 Mod 0
DL 2645361	Shipping Arrangement
LD 166537	Test Set Mk 135 Mod 0
LD 255258	Primer Container, Mk 11 Mod 4
512849	Primer Container, Mk 7 Mods 0 & 1
553214	Firing Pin for Primers
2645361	Shipping Condition for Primers Mk 41 and Mk 42
2645488	Palletizing Domestic Unit Load
2839128	Cartridge Assembly, 3"/50 BL & P

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

WR 53/819	Palletizing Domestic Unit Load
WR 53/851	Palletizing Domestic Unit Load
OD 5823	Sensitivity Test of Primers and Detonators
NAVSEANOTE 4855	Classification of Characteristics

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

CFR Title 49 - Transportation (Parts 100 to 199)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

3. REQUIREMENTS

3.1 Item definition. The primer consists of a threaded stock containing a percussion primer M61 which contains an FA-70 primer ignition mix, connected to a seamless steel extension tube containing class 2 black powder. The individual components of the primer are listed on DL 563478.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Function. The primer shall meet the following requirements:

a. (C1) After firing, the primer shall not exhibit a split stock, split tube or detachment of metallic components.

b. (M101) Primer functioning time shall have an upper specification limit of .006 seconds.

c. (M102) The primer shall not misfire (see 6.5.1).

d. (M103) The primer shall not exhibit blowback (see 6.5.2).

3.2.1.2 Ballistic. The primer shall meet the following requirements:

a. (C2) After firing, the primer shall not exhibit a split stock, split tube or detachment of metallic components.

b. (M104) The primer shall fire, and the projectile shall be ejected from the gun barrel within .050 seconds after application of firing energy.

c. (M105) The initial velocity standard deviation shall be sixteen feet per second maximum.

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d. (M106) The primer shall not exhibit blowback (see 6.5.2).

e. (M107) The primer shall not misfire (see 6.5.1).

3.2.1.3 Vibration. The primer shall meet the following requirements:

a. (C3) The primer shall not fire.

b. (C4) The stock and tube shall maintain material integrity and the primer shall maintain assembly integrity at the mating threads of the stock and tube.

c. The primer shall meet the requirements of 3.2.1.1 after vibration.

3.3 Design and construction.

3.3.1 Construction. The primer shall conform to all the requirements specified herein and specified in all the documents listed on DL 563478.

3.3.1.1 Radiographic. The primer shall not exhibit any loose (except black powder), missing, or extra components. Primers exhibiting possible defects shall be included in detail strip sample and meet the requirements of 3.3.1.2.

3.3.1.2 Detail strip. The primer and all its components shall conform to the following requirements specified on the applicable documents listed on DL 563478 when examined in accordance with 4.6.5:

a. (C5) Each critical characteristic; and

b. (M108) Each major characteristic.

3.3.2 No-fire. (C6) The primer shall not fire when tested in accordance with 4.6.6.

3.3.3 Workmanship. All parts shall be free of chips, dirt, grease, rust, and all other foreign materials. There shall be no burrs or other imperfections of manufacture which would harmfully affect the assembly, functioning or handling of the product. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

3.4 First article sample. Unless otherwise specified in the contract (see 6.2.1), a First Article shall be manufactured, inspected, and accepted prior to production. Component inspection shall be part of that requirement for all First Articles. The Government may also require at

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its discretion, the manufacture, inspection, and acceptance of a First Article if any of the following conditions exist:

a. There has been a break in production of at least six months duration.

b. The supplier has made a major change in his assembly procedures.

c. There has been a major change in production personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract (see 6.2.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may utilize his own or any other facilities acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article. The intent of the First Article is to demonstrate that, prior to production, the producer's fabrication methods are capable of yielding primers that comply with the technical requirements of the contract.

4.3.1 Formation. The First Article shall be a homogeneous sample of 236 primers fabricated, packaged, and packed in accordance with the requirements of this specification and the contract using the methods and procedures to be used for production.

4.3.2 Inspection. The First Article shall be submitted to the facility designated in the contract (see 6.2.1) for test and examination in accordance with the first article inspection sampling plan of TABLE I.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Unless otherwise specified in the contract (see 6.2.1), the production lot shall consist of 10,800 (maximum) loaded primers plus those required for test purposes. Each production lot shall contain only one M61 percussion primer lot and only one black powder lot.

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TABLE I. First article and quality conformance inspection and sampling plans.

Test or Examination	Requirement Paragraph	Method Paragraph	First article inspection			Quality conformance inspection		
			Sample Size	Acceptance Criteria		Sample Size	Acceptance criteria	
				Accept	Reject		Accept	Reject
Radiographic [1]	3.3.1.1	4.6.4	236[1]	0	Go to 3.3.1.2 (Detail Strip)	151[1]	0	Go to 3.3.1.2 (Detail Strip)
Function a. (C1) b. (M101) c. (M102)(M103)	3.2.1.1	4.6.1	MIL-STD-414 Sample Size Code ltr 0	0 AQL≤1%	1 AQL>1%	MIL-STD-414 Sample Size Code ltr N	0 AQL≤1%	1 AQL>1%
Ballistic a. (C2) b. (M104)(M106) (M107) c. (M105)	3.2.1.2	4.6.2	20	0 Combined sample must meet 3.2.1.2.C	1			
Vibration a. (C3)(C4) b. (M101) c. (M102)(M103) (C1)	3.2.1.3 3.2.1.1	4.6.3 4.6.1	30	0	1	20	0	1
Detail strip a. (C5) b. (M108)	3.3.1.2	4.6.5	56 30[2]	0 1	1 2	56 30[2]	0 1	1 2
No-fire (C6)	3.3.2	4.6.6	56[3]	0	1	56[3]	0	1
Packaging	5.1	4.6.7	1	0	1			
Packing	5.2	4.6.7	1	0	1			
Marking (M109)	5.4	4.6.7	All marking	0	1			

[1] Radiographic examination of each primer is to be performed prior to any other test or examination.
[2] Samples are same as for a. above. Acceptance criteria is based on the number of defects for each major characteristic.
[3] Samples are same as for Detail Strip examination.

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4.4.2 Inspection. Unless otherwise specified in the contract (see 6.2.1), 151 samples shall be selected at random from each production lot for test and examination in accordance with the quality conformance inspection plan of TABLE I.

4.5 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as critical, major or minor in accordance with NAVSEANOTE 4855. Critical test and examination characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Test and examinations which are not annotated with a classification code are classified minor.

4.6 Test and examination methods. Government approval of all test procedures and equipment is required prior to their use for acceptance purposes (see 6.2 and 6.4). Primers shall be selected at random for each test or examination. Identification of each primer's complete test and examination results shall be maintained. Any primer (or part thereof) exhibiting a defect shall be secured for additional examination until otherwise directed by the procuring activity. A defective is any primer which fails to meet the associated requirements for the examination and test described in TABLE I (see 4.3.2).

4.6.1 Function. The primer shall be assembled into test set Mk 135 Mod 0 and tested in accordance with the appropriate procedures of OD 5823 using appropriate test equipment including the following:

a. Firing pin; Dwg. 553214, piece 4.

b. Steel ball; approximately 1 29/32 inch diameter, 16.34 ± .02 ounces.

The ball drop height shall be 12 inches. Firing time is the time from first application of firing energy to the primer until the first indication of light is produced. First indication of light shall be determined by use of two photo transistors or light activated silicon controlled rectifiers aimed parallel to the primer tube and located at the stock end of the primer.

4.6.2 Ballistic. The primer shall be assembled into 3"/50 cartridges in accordance with Dwg. 2839128. All components shall be of service acceptable lots. The propellant shall be conditioned at 90° ± 20°F for 24 hours minimum prior to assembly of the propelling charge. All other components shall be conditioned at 85° ± 20°F for 24 hours minimum prior to assembly of the propelling charge. The assembled propelling charge shall be conditioned at 90° ± 5°F for 48 hours minimum immediately prior to firing. The rounds shall be fired in 3"/50 percussion fired gun with a first quarter life barrel. Initial projectile muzzle velocity, projectile ejection time, and peak pressure at the base of the cartridge case shall be recorded.

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Initial velocity standard deviation, s, shall be calculated as follows:

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{N - 1}}$$

where:

s = initial velocity standard deviation, ft/sec
X = velocity of each firing, ft/sec
N = number of firings, and

$\bar{X}$ = mean velocity = $\frac{\Sigma X}{N}$, ft/sec

4.6.3 Vibration. The primer shall be subjected to the transportation vibration test of MIL-STD-331, Test 104, Procedure II, except that the orientation of each primer during this three phase test shall be as follows:

a. Phase I: Primer axis horizontal;

b. Phase II: Primer axis vertical, primer stock down; and,

c. Phase III: Primer axis vertical, primer stock up.

The primer shall be rigidly mounted to the vibration table at the stock threads.

4.6.4 Radiographic. The primer shall be subjected to a radiographic inspection in accordance with MIL-STD-453. The primer shall be mounted in the vertical position with the primer tube down, and x-rayed perpendicular to its axis. Any primer which appears to have been misassembled shall be subjected to detail strip and shall meet the requirements of 3.3.1.2.

4.6.5 Detail strip. The primer shall be disassembled, except for the primer stock assembly (see 6.5.3) to determine conformance to all the discernible critical and major characteristics specified on the applicable documents listed on DL 563478. All inspections shall be performed at the highest level of assembly possible. The primer stock shall not be mishandled during examination to the extent that its performance will be affected (see 4.6.6).

4.6.6 No-fire. The primer stock assembly from 4.6.5 shall be assembled into test set Mk 135 Mod 0 (LD 166537) and tested in accordance with the appropriate procedures of OD 5823 using the test equipment specified in 4.6.1. The ball drop height shall be three inches.

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4.6.7 Packaging, packing, and marking. The packaging, packing, and marking for the primer shall be visually examined for conformance to the major requirements of Section 5.

5. PACKAGING

5.1 Packaging. The primer shall be packaged in accordance with DL 2645361 (preferred) or LD 255258 (alternate). All requirements of these documents and their classification apply.

5.1.1 Inert components. Components and subassemblies shall be packaged in commercial fiberboard containers meeting the requirements of PPP-B-636. Each container shall incorporate corrugated or solid fiberboard dividers or collars suitable for preventing damage in handling and transportation. The gross weight of the container contents shall not be greater than 70 pounds.

5.2 Packing. Primers packaged in accordance with 5.1 shall be packed in accordance with Dwg. 2645361 (preferred) or Dwg. 512849 (alternate) and as specified in the contract (see 6.2). All requirements of these documents and their classification apply.

5.3 Palletization. Palletization of primers packed in accordance with 5.2 shall be in accordance with Dwg. 2645488 for preferred method or WR-53/819 or WR-53/851 as applicable for alternate method.

5.4 Marking. (M109) In addition to any special marking required by the contract, all markings shall be in accordance with MIL-STD-129 and CFR Title 49 - Transportation (Parts 100 - 199)

5.5 Data cards. Data cards shall be furnished in accordance with MIL-STD-1167. The quantity of cards supplied shall be as specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The primer Mk 41 Mod 0 is intended for use as an initiator in 3"/50 caliber percussion fired ammunition.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Whether a first article sample is required (see 3.4).

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c. The facility designated to perform inspections (see 4.1).

d. Inspection lot quantity(ies) if other than as specified in 4.4.1.

e. Inspection lot sample quantity if other than as specified in 4.4.2.

f. Safety precautions (see 6.3).

g. Whether primers are to be packed using the preferred or the alternate methods (see 5.2).

h. The quantity of ammunition data cards to be supplied (see 5.5).

6.2.2 Contract data requirements. The items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.6	Test Procedure	UDI-T-22710
5.5	Ammunition Data Card	DI-E-2001

DID's (Data Item Descriptions/DD Forms 1664) for the above data requirements are documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DID's when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Safety precautions. The safety precautions requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.4 Test procedures and reporting. For Navy procurements, the procuring activity should include in each work authorization for first article and production inspections, the requirement for the preparation and forwarding to the activities listed below of all test and examination procedures prior to performance of any inspection. The work authorization should also include the requirements for the forwarding of all test data in an intelligible form to the activities listed below within 10 days of the completion of the tests.

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Naval Weapons Support Center
Crane, Indiana 47522

Commanding Officer
Naval Ordnance Station (5231)
Indian Head, Maryland 20640

6.5 Definitions.

6.5.1 Misfire. Misfire is the failure of the main black powder charge of the primer to ignite when the primer is subjected to the appropriate firing source.

6.5.2 Blowback. Blowback is gas leakage around or through the stock end of the primer and is verified by observance during firing and/or by visual evidence of residue after firing.

6.5.3 Primer stock assembly. The primer stock assembly consists of the primer stock, M16 percussion primer, battery cup, and firing plug.

6.6 Changes from previous issue. Due to the comprehensive nature of this revision the margins of this specification have not been marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-18515B, PRIMER, PERCUSSION, MK 41 MOD 0

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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